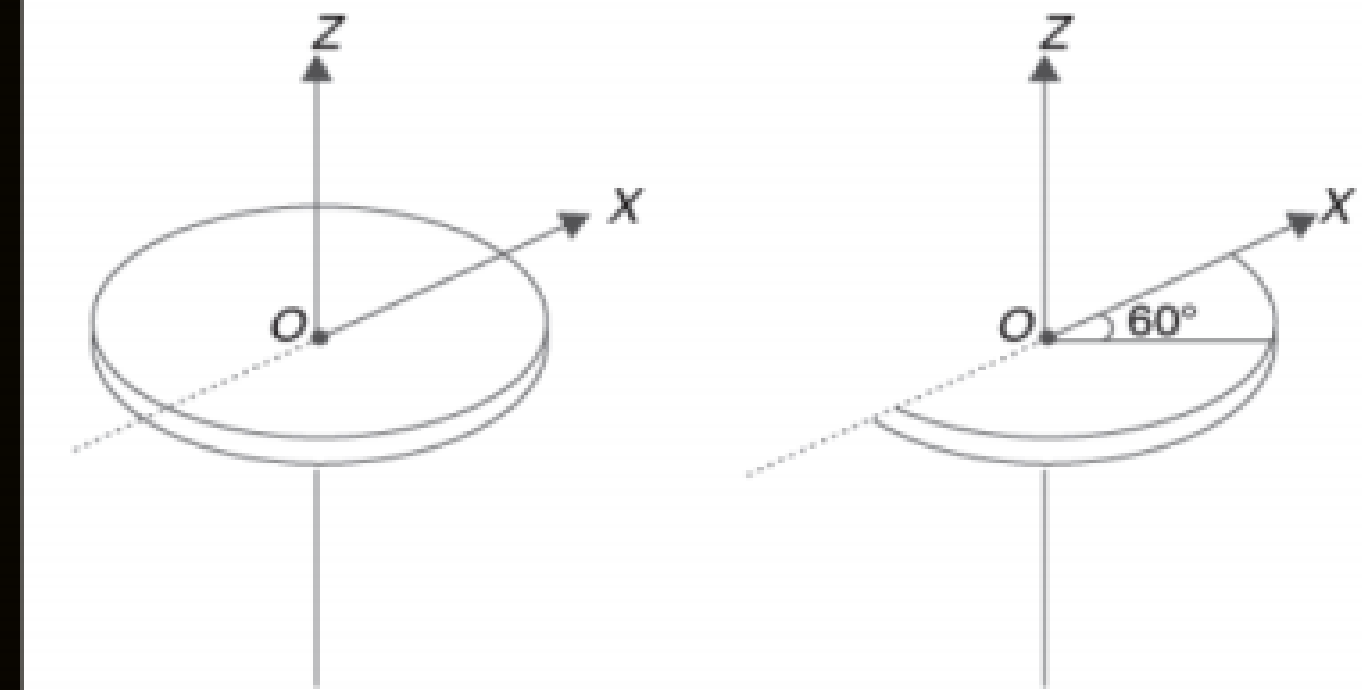




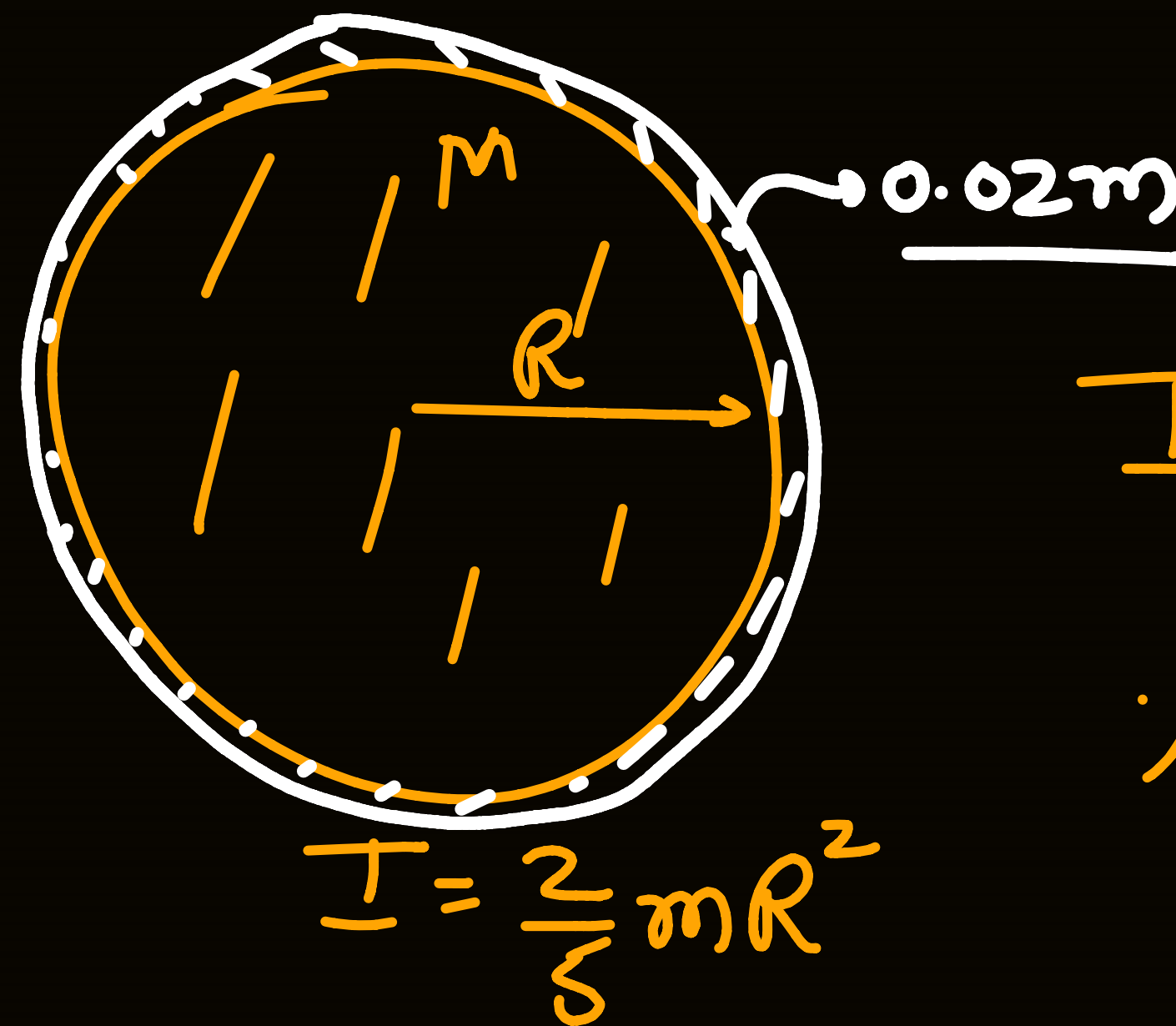
An Indian bread "Roti" is a uniform disc of mass M and radius R . Before eating a person usually folds it about its diameter (say about x axis). After folding it a sector of angle 60° is removed from it. Find the moment of inertia of the remaining "Roti" about Z -axis.



$$3I = \frac{MR^2}{2}$$

$$2I = \frac{\frac{MR^2}{2} \times 2}{3} = \frac{MR^2}{3}$$

An asteroid in the shape of a uniform sphere encounters cosmic dust. A thin uniform layer of dust gets deposited on it and its mass increases by 2%. Find percentage change in its moment of inertia about diameter.



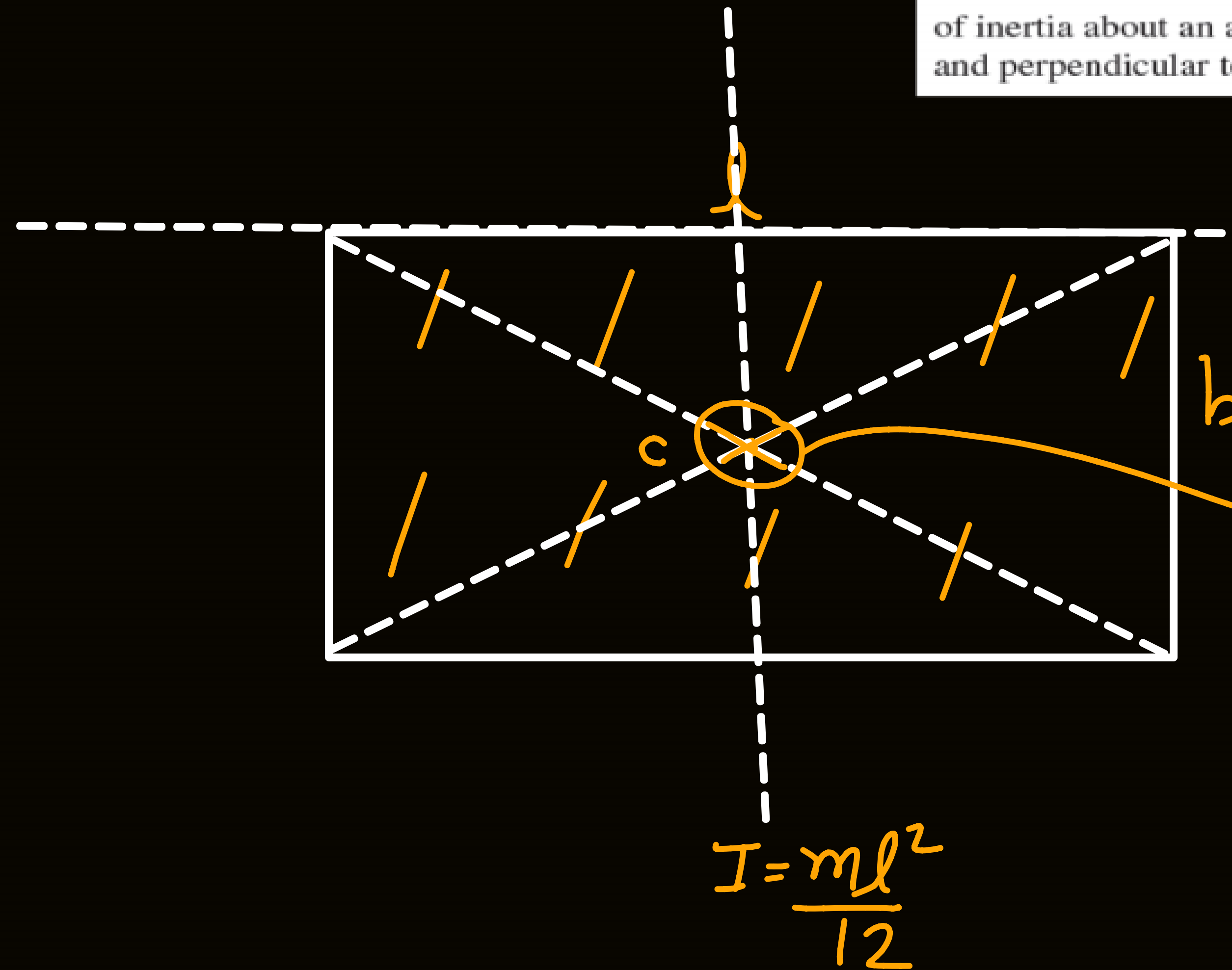
$$I' = \frac{2}{5}mR^2 + \frac{2}{3}(0.02m)R'^2$$

$$\begin{aligned} \therefore \text{Change} &= \frac{I' - I}{I} \times 100 = \frac{\frac{2}{3} \times 0.02mR'^2}{\frac{2}{5}mR^2} \times 100 \\ &= \frac{10}{3} \% \end{aligned}$$

A uniform rectangular plate has moment of inertia about its longer side, equal to I . The moment of inertia of the plate about an axis in its plane, passing through the centre and parallel to the shorter sides is also equal to I . Find its moment of inertia about an axis passing through its centre and perpendicular to its plane.

$$\frac{ml^2}{4} = \frac{mb^2}{3}$$

$$\boxed{l = 2b}$$

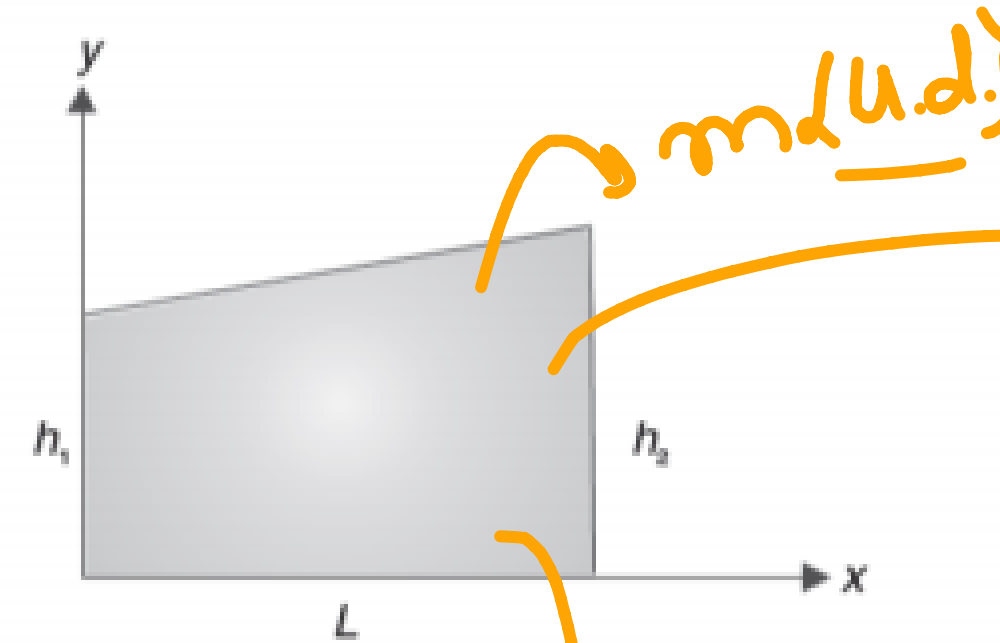


$$I' = \frac{ml^2}{12} + \frac{mb^2}{12}$$

$$= I + \frac{m}{12} \cdot \frac{l^2}{4}$$

$$= I + I/4 = 5I/4$$

Determine the moment of inertia of the shaded area about y axis. The mass of the shaded area is M .



$$\sigma = \frac{2m}{(h_1 + h_2)L}$$

$$Area = \frac{1}{2}(h_1 + h_2)L$$

$$y = h_1 + \frac{h_2 - h_1}{L}x$$

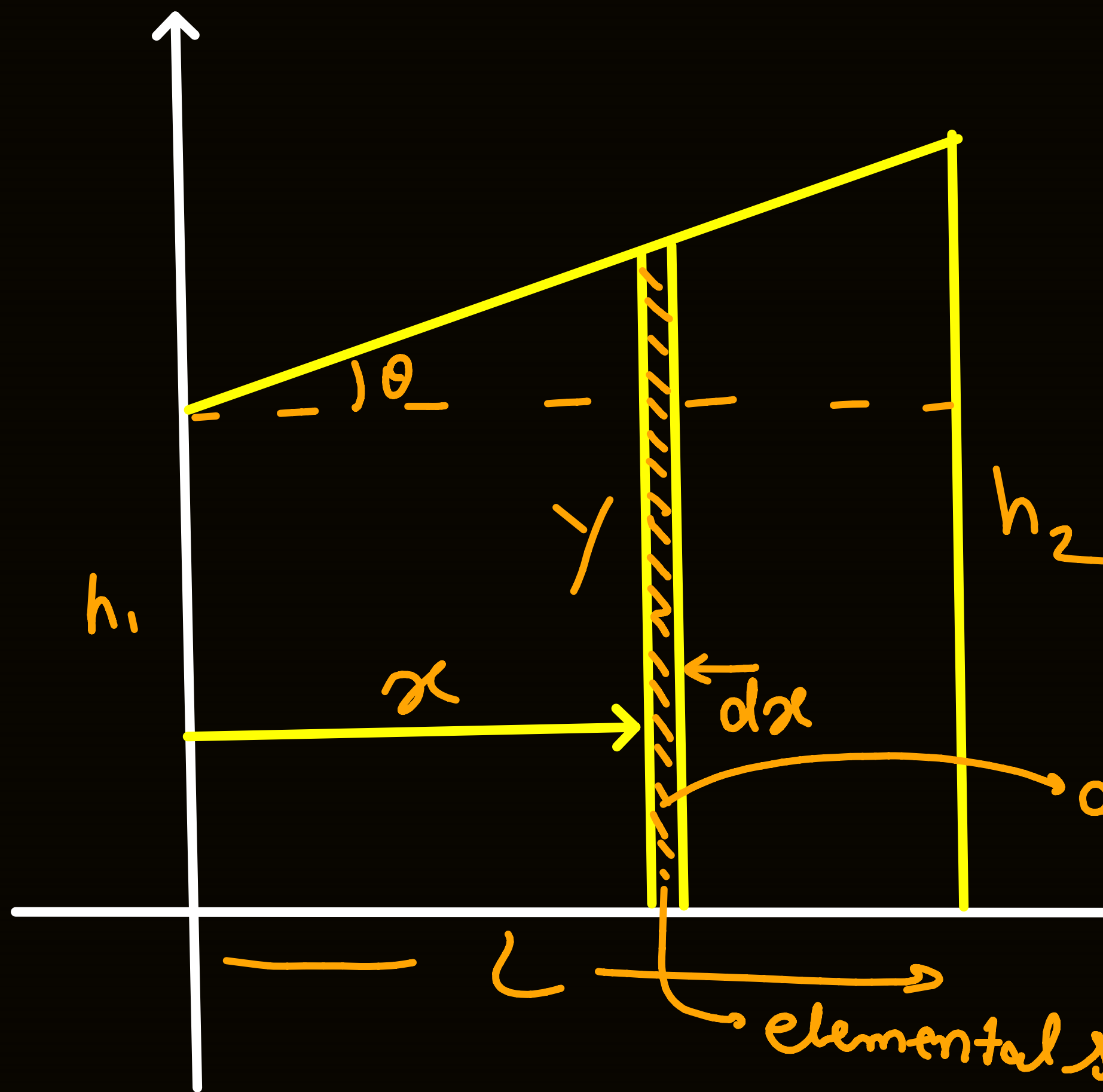
$$dm = \sigma dA = \frac{2m}{(h_1 + h_2)L} \left\{ h_1 + \frac{h_2 - h_1}{L}x \right\} dx$$

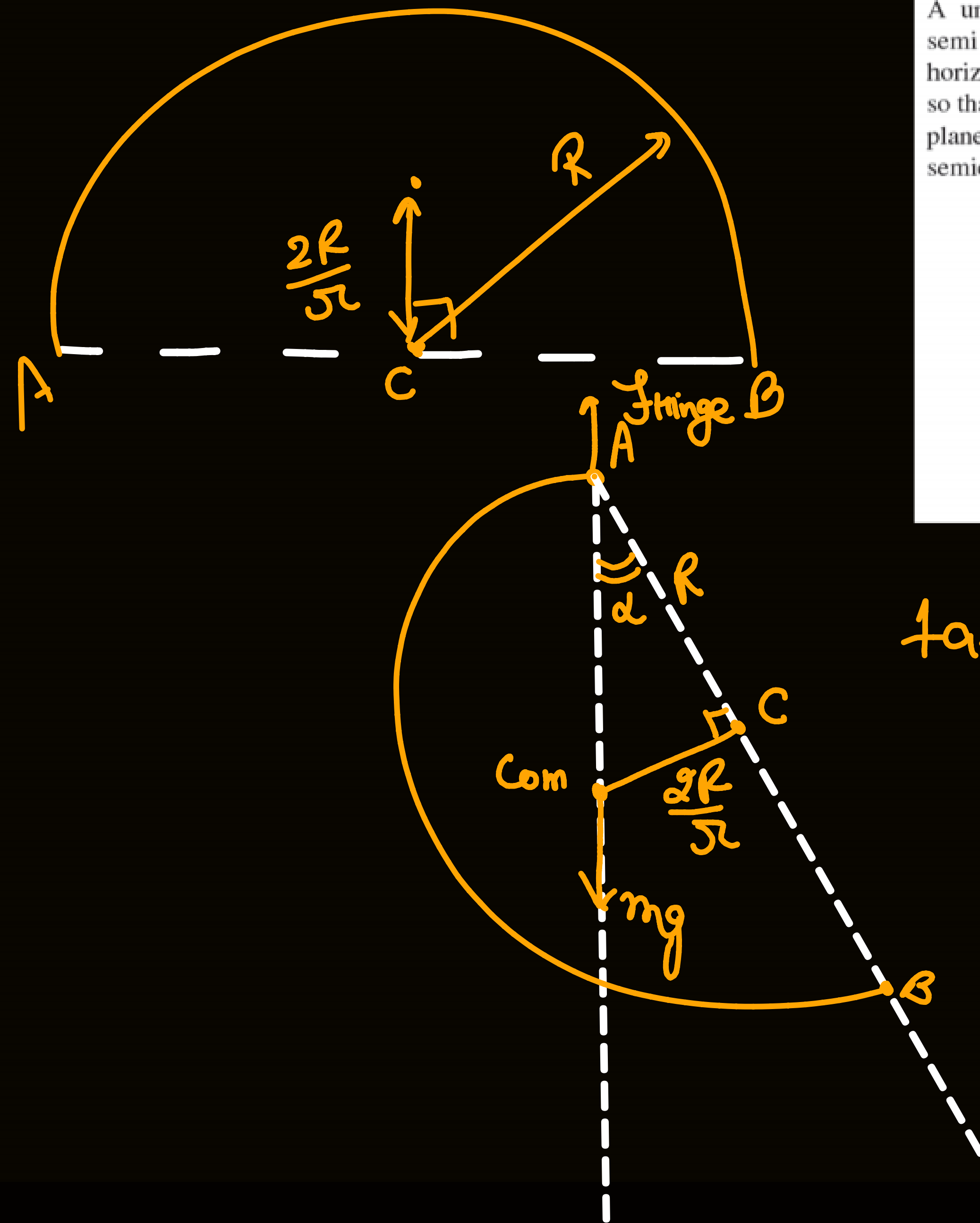
$$I = \int dI = \int (dm) x^2$$

$$= \frac{2m}{(h_1 + h_2)L} \left\{ h_1 \int_0^L x^2 dx + \frac{h_2 - h_1}{L} \int_0^L x^3 dx \right\}$$

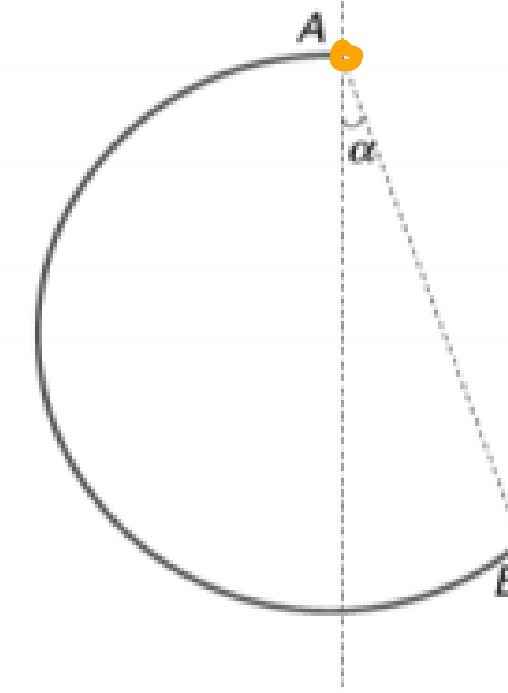
$$\tan \theta = \frac{h_2 - h_1}{L} = \frac{y - h_1}{x}$$

elemental strip





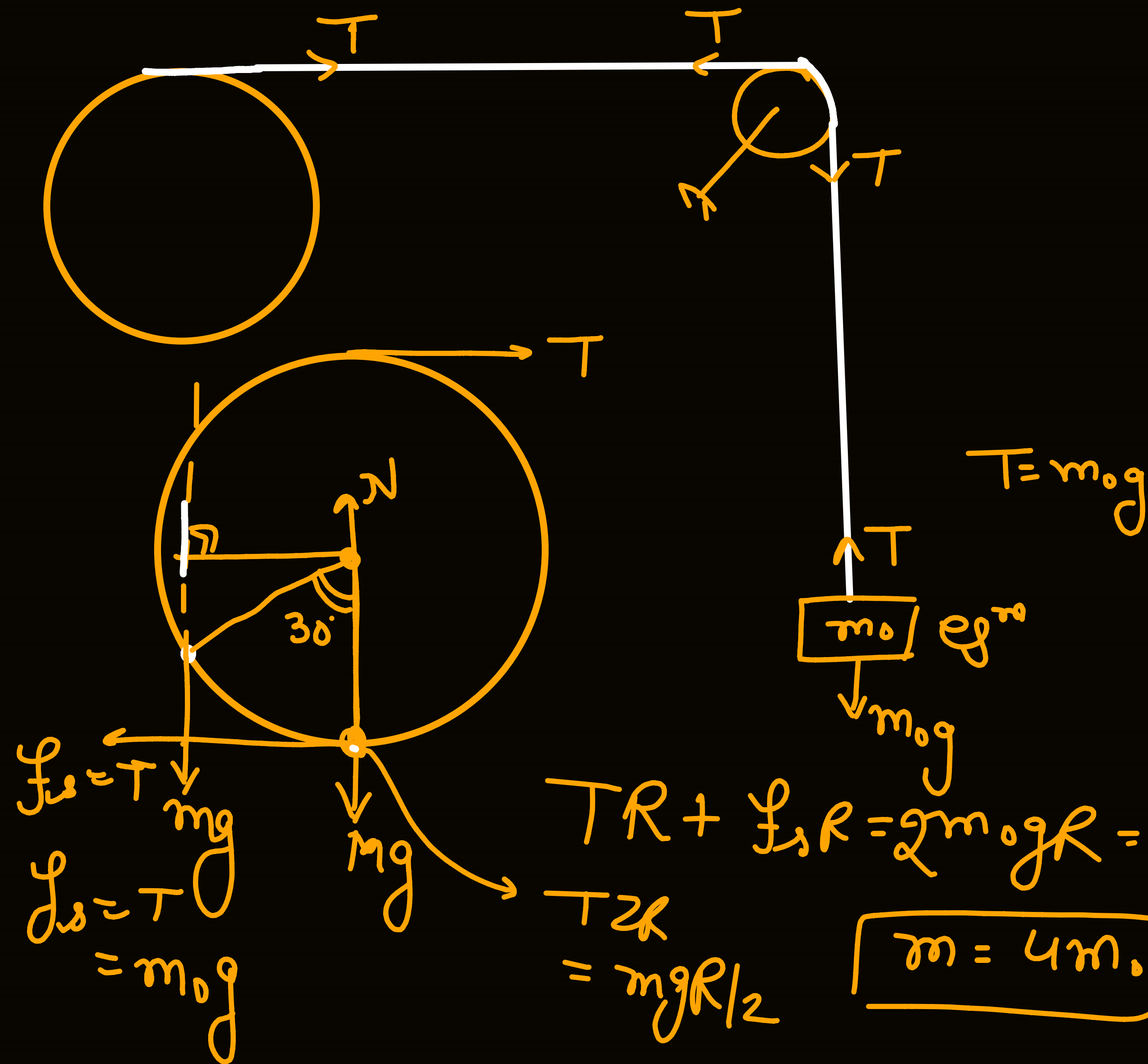
A uniform wire has been bent in shape of a semi circle. The semicircle is suspended about a horizontal axis passing through one of its ends, so that the semicircular wire can swing in vertical plane. Find the angle α that the diameter of the semicircle makes with vertical in equilibrium.



$$\tan \alpha = \frac{2R/5}{R}$$

$$= 2/5$$

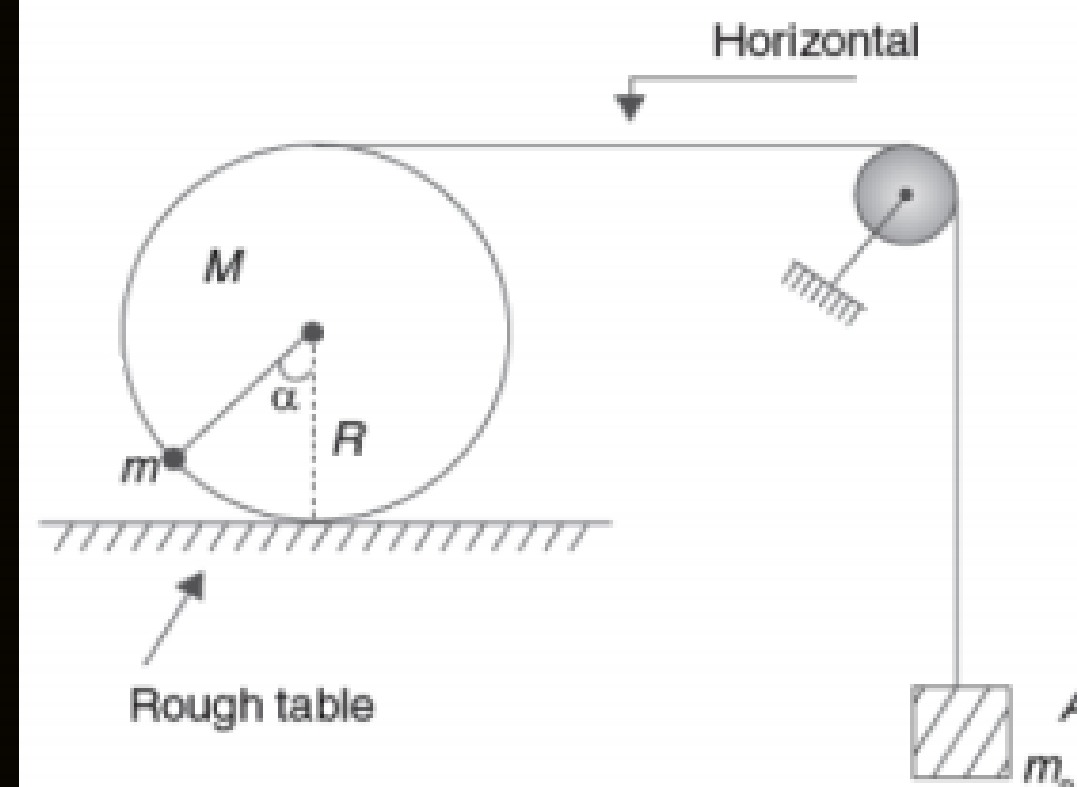
$$\alpha = \tan^{-1}\{2/5\}$$

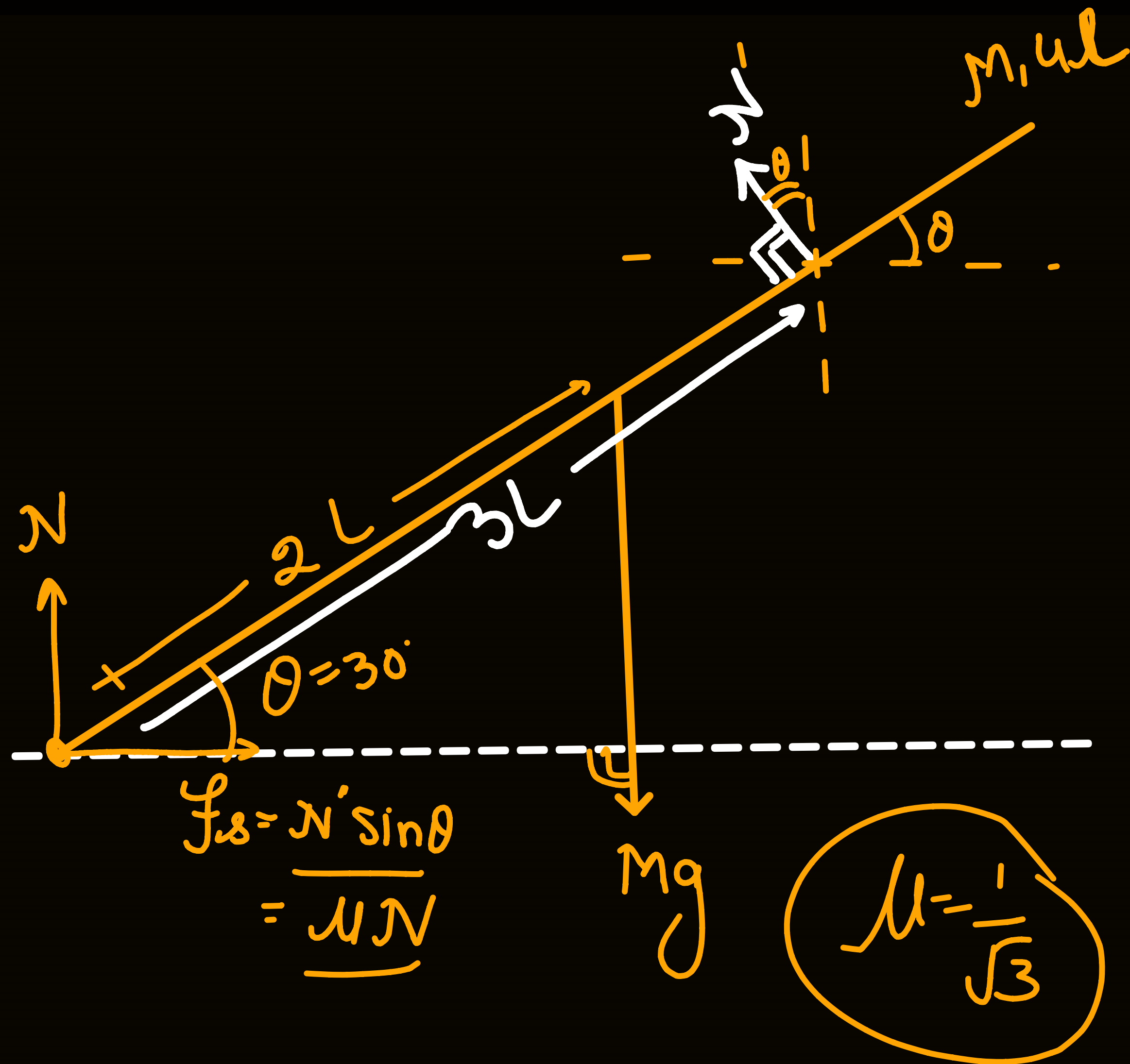


A hollow cylindrical pipe of mass M and radius R has a thin rod of mass m welded inside it, along its length. A light thread is tightly wound on the surface of the pipe. A mass m_0 is attached to the end of the thread as shown in figure. The system stays in equilibrium when the cylinder is placed such that $\alpha = 30^\circ$. The pulley shown in figure is a disc of mass $\frac{M}{2}$.

(a) Find the direction and magnitude of friction force acting on the cylinder.

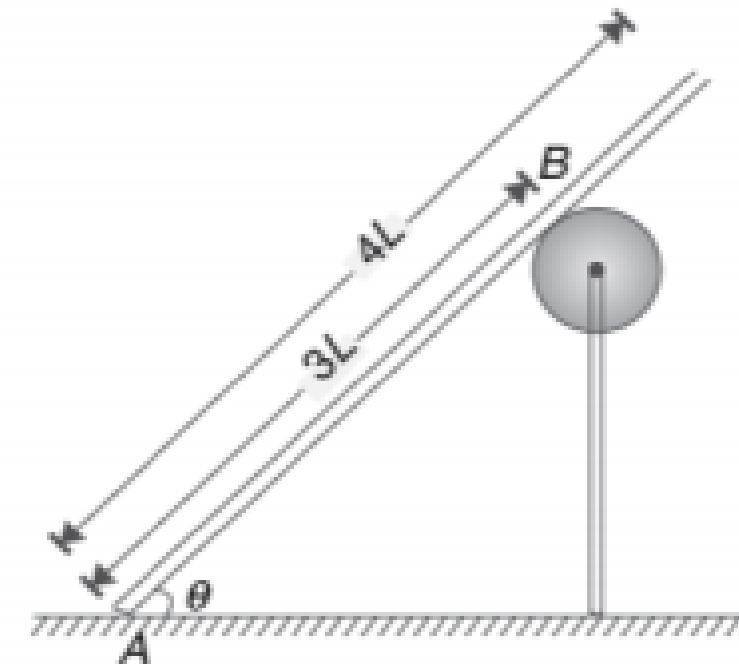
(b) Express mass of the rod ' m ' in terms of m_0





A uniform rod has mass M and length $4L$. It rests in equilibrium with one end on a rough horizontal surface at A. At point B, at a distance $3L$ from A, it is supported by a fixed smooth roller. The rod just remains in equilibrium when $\theta = 30^\circ$

- Find the normal force applied by the horizontal surface on the rod at point A.
- Find the coefficient of friction between the rod and the surface.



$$f_s = \frac{N'}{2} \quad (2)$$

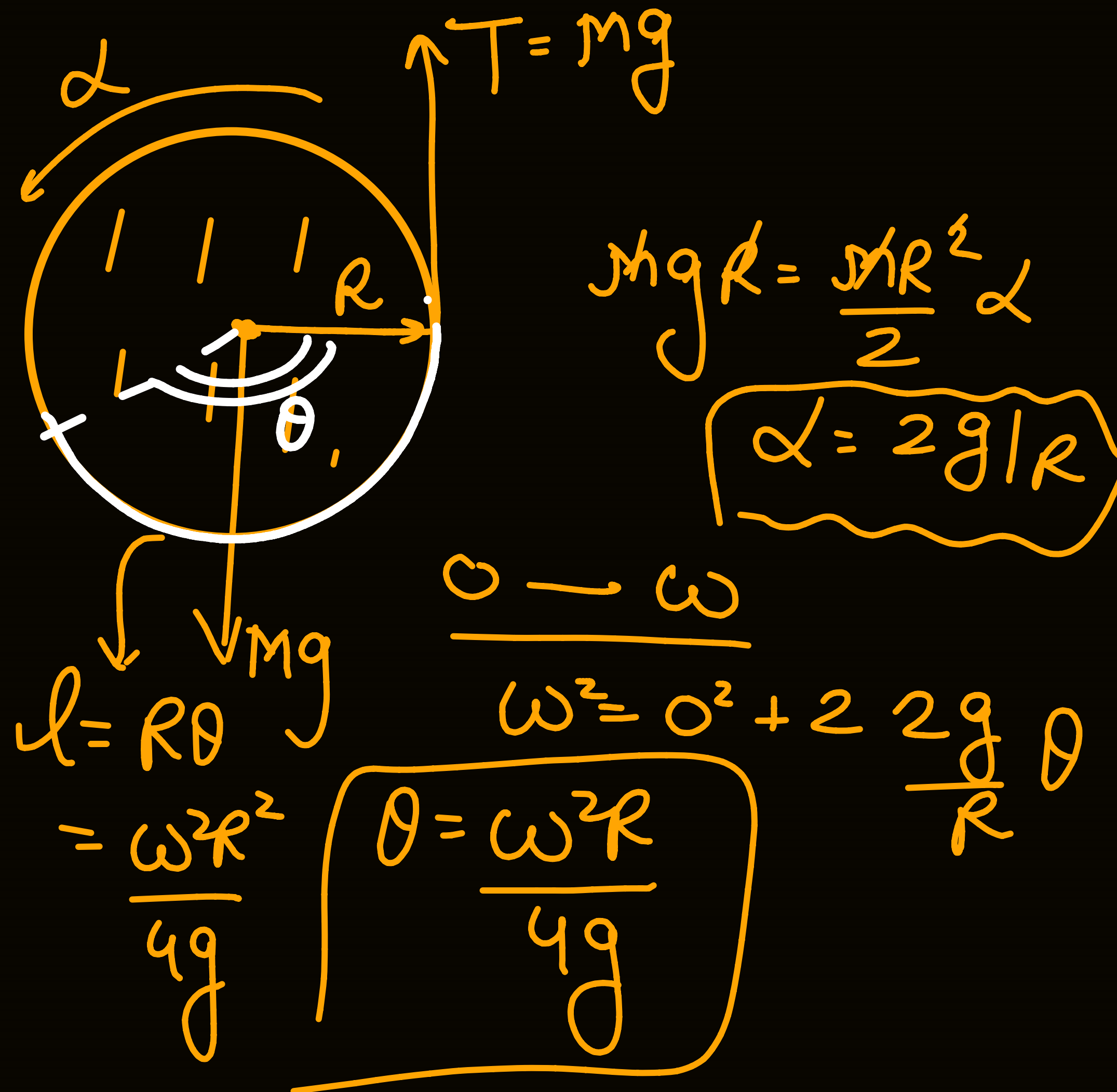
$$f_s = \frac{Mg}{2\sqrt{3}} = \mu \frac{Mg}{2}$$

$$N + \frac{N'\sqrt{3}}{2} = Mg \quad (1)$$

$$Mg \cdot 2L \frac{\sqrt{3}}{2} = N' \cdot 3L \quad (3)$$

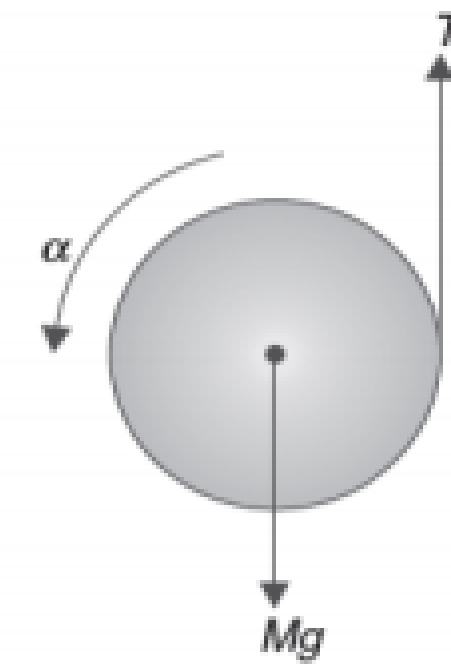
$$N' = \frac{Mg}{\sqrt{3}}$$

$$N = Mg - \frac{Mg}{2} = \frac{Mg}{2}$$

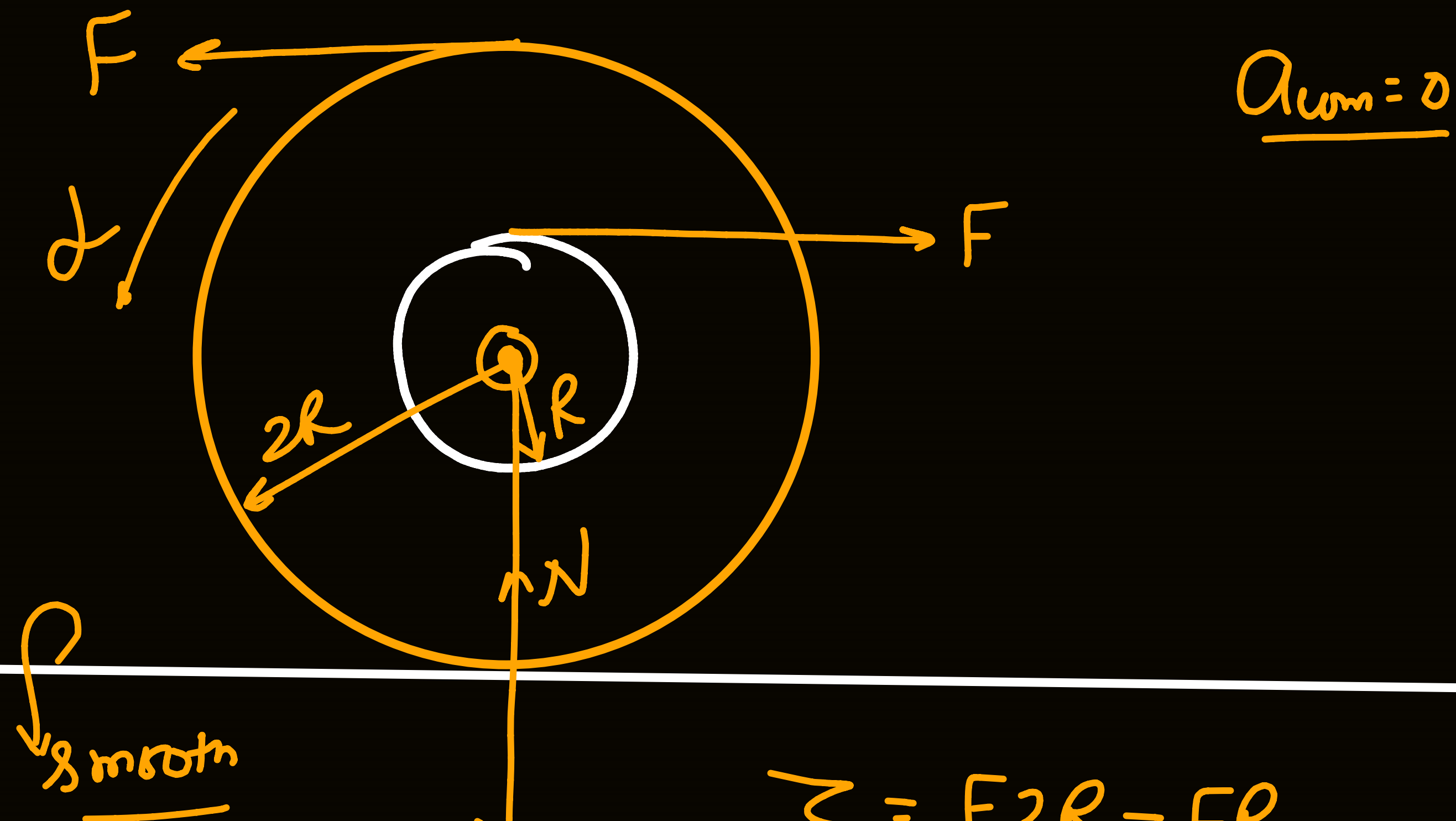


$T = Mg$
 $MgR = \frac{MR^2}{2} \alpha$
 $\alpha = \frac{2g}{R}$
 $0 \rightarrow \omega$
 $\omega^2 = 0^2 + 2 \cdot \frac{2g}{R} \theta$
 $\theta = \frac{\omega^2 R}{4g}$
 $l = R\theta = \frac{\omega^2 R^2}{4g}$

A string is wrapped around a cylinder of mass M and radius R . The string is pulled vertically upward to prevent the centre of mass from falling as the string unwinds. Assume that the cylinder remains horizontal throughout and the thread does not slip. Find the length of the string unwound when the cylinder has reached an angular speed ω .



①



8 m/s²

$$\omega = 0 + \frac{5}{2} \times 2$$

$$= 5$$

$$K.E. = R.K.E. = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times 8 \times 25 = 100 \text{ J.}$$

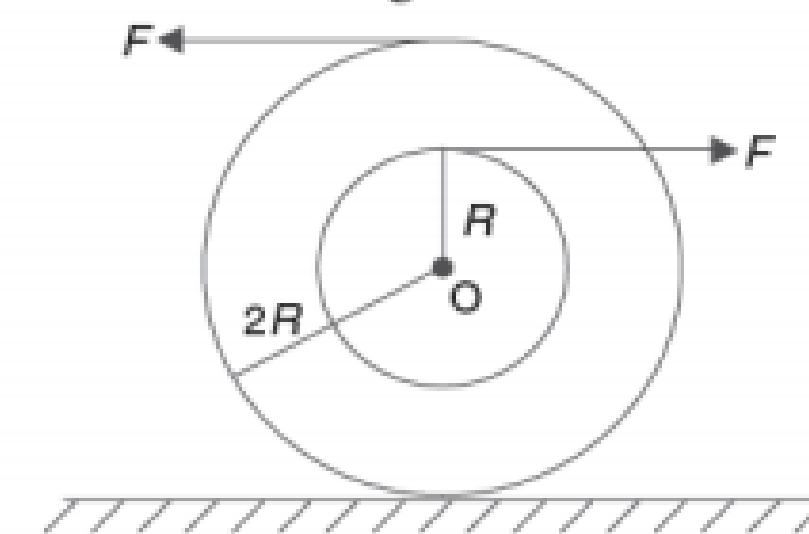
$$\tau = F2R - FR$$

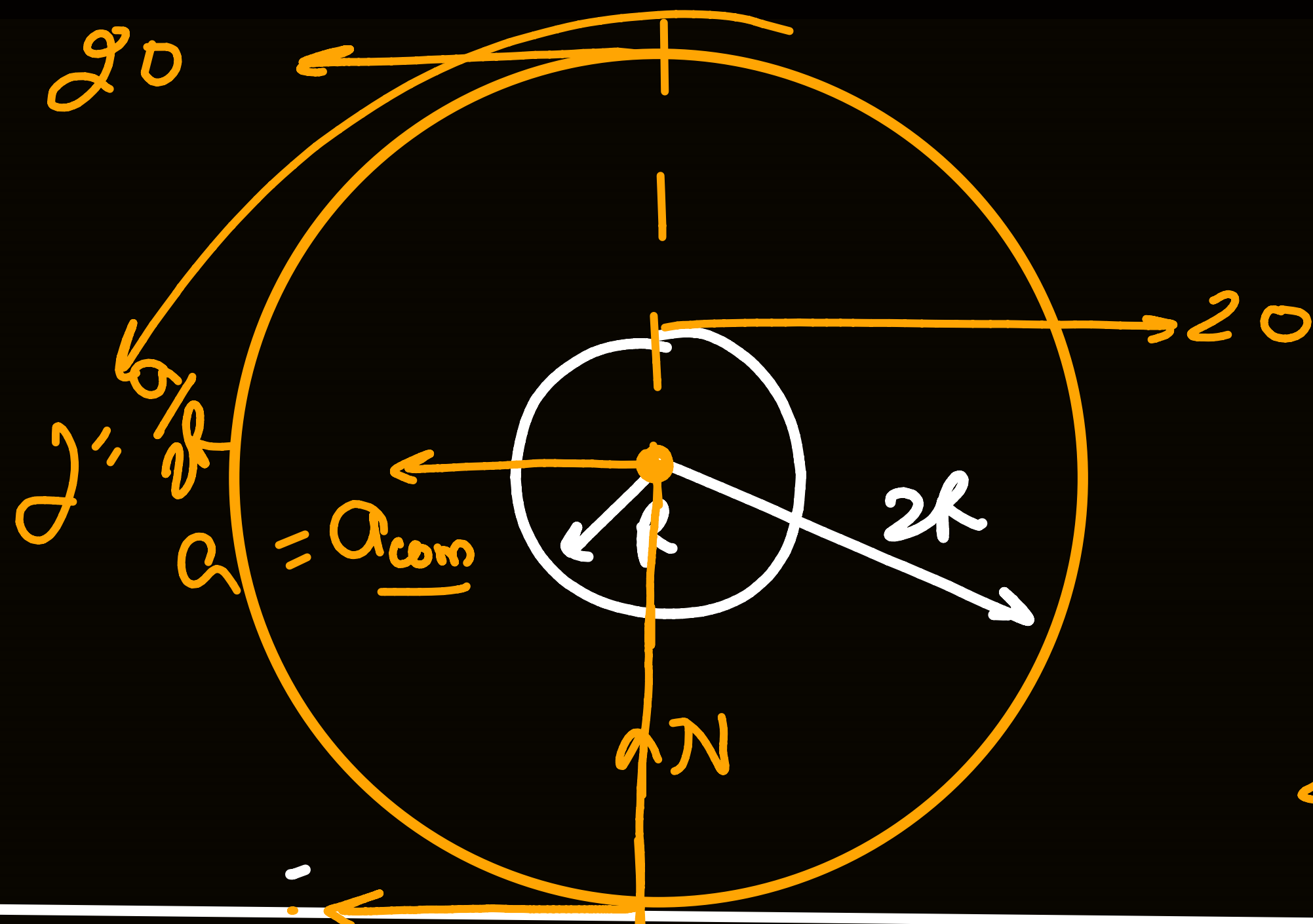
$$= FR = I\alpha$$

$$\alpha = \frac{FR}{I} = \frac{20 \times 1}{8} = 2.5 \text{ rad/s}^2$$

A disc shaped body has two tight windings of light threads - one on the inner rim of radius $R = 1\text{m}$ and the other on outer rim of radius $2R$ (see figure). It is kept on a horizontal surface and the ends of the two threads are pulled horizontally in opposite directions with force of equal magnitude $F = 20\text{N}$. Mass of the body and its moment of inertia about an axis through centre O and perpendicular to the plane of the figure are $M = 4\text{kg}$ and $I = 8\text{kg} - \text{m}^2$ respectively. Find the kinetic energy of the body 2 seconds after the forces begin to act, if

- the surface is smooth,
- the surface is rough enough to ensure rolling without sliding.





$$f_s = 4a \quad \text{--- (1)}$$

$$20 \times 2 - 20 \times 1$$

$$- f_s \times 2 = 8a$$

$$20 - 2f_s = 4a \quad \text{--- (2)}$$

$$20 - 8a = 4a$$

$$12a = 20$$

$$a = \frac{5}{3} \text{ m/s}^2$$

$$\alpha = \frac{a}{2R} = \frac{5}{6} \text{ rad/s}^2$$

$$v_{\text{com}} = 0 + \frac{5}{3} \times 2$$

$$= \frac{10}{3} \text{ m/s}$$

$$\omega = 0 + \frac{5}{6} \times 2 = \frac{5}{3} \text{ rad/s}$$

$$K.E = \frac{1}{2} m v_{\text{com}}^2 + \frac{1}{2} I_c \omega^2$$

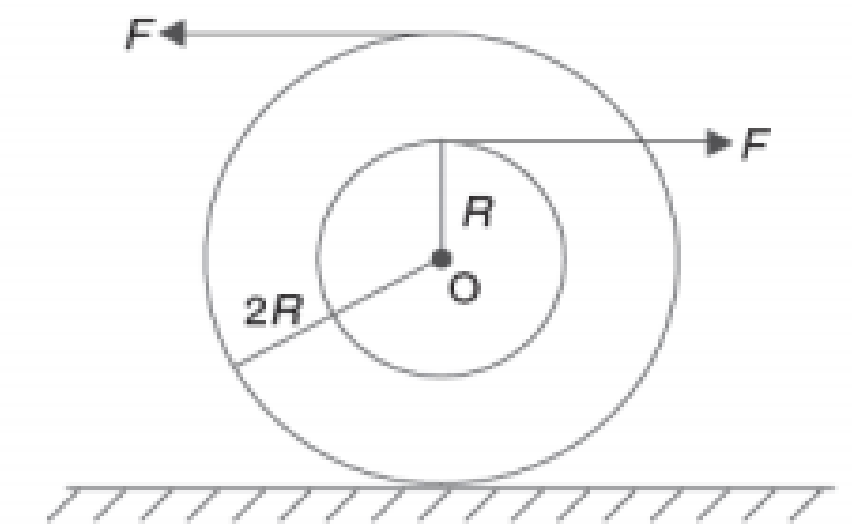
$$= \frac{1}{2} \times 4 \times \left(\frac{10}{3}\right)^2 + \frac{1}{2} \times 8 \times \left(\frac{5}{3}\right)^2$$

$$= \frac{200}{9} + \frac{100}{9} = \frac{300}{9} \text{ Joules}$$

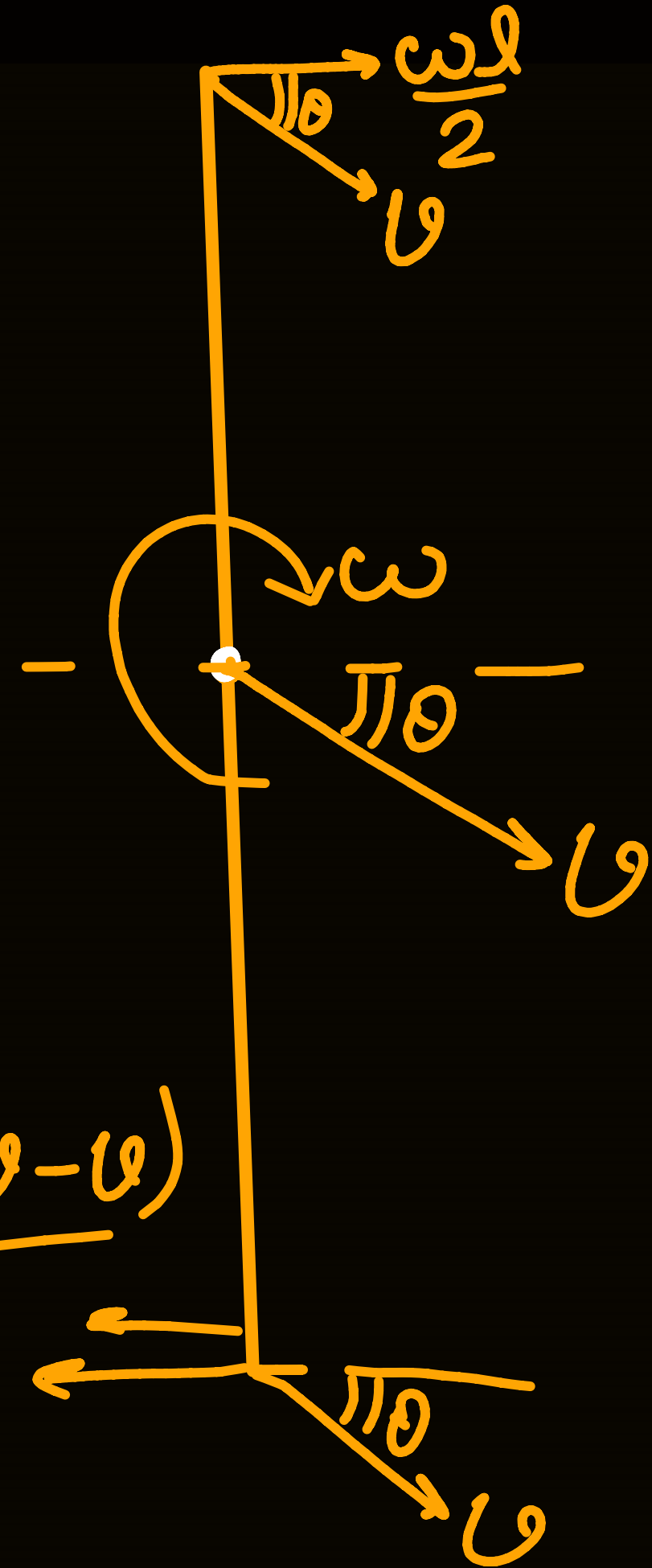
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(i) the surface is smooth,

(ii) the surface is rough enough to ensure rolling without sliding.



$$KE = TKE + RKE = \frac{1}{2} mV^2 + \frac{1}{2} \frac{ml^2}{12} \omega^2 = \frac{4}{3} mV^2$$



$$\sqrt{5}V = \frac{\omega l}{2}$$

$$V_{\max} = V + \frac{\omega l}{2}$$

$$= \boxed{V + \sqrt{5}V}$$

$$\frac{1}{2} \frac{ml^2}{12} \omega^2 = \left(\frac{4}{3} - \frac{1}{2} \right) mV^2$$

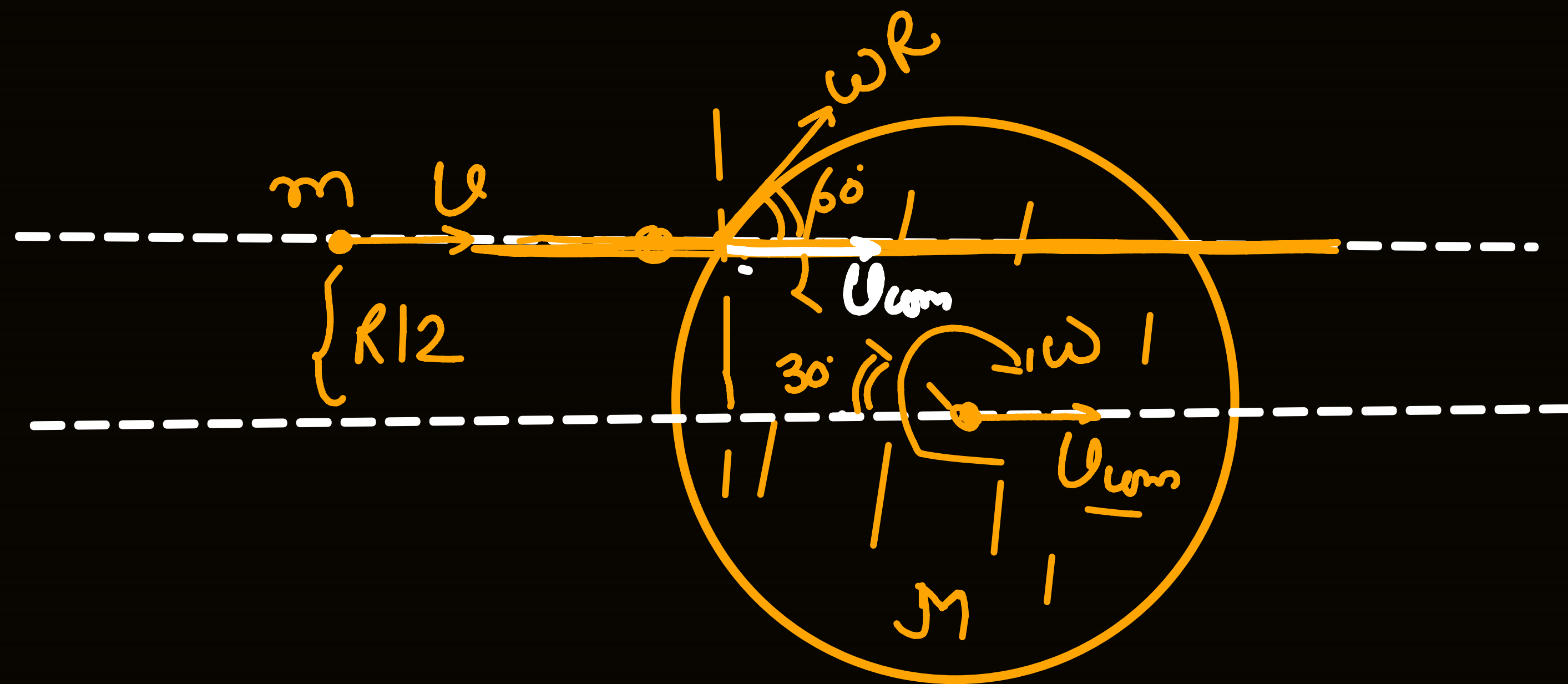
$$= \frac{5}{6} mV^2$$

$$l^2 \omega^2 = 20V^2$$

$$\boxed{\omega l = \sqrt{20}V}$$

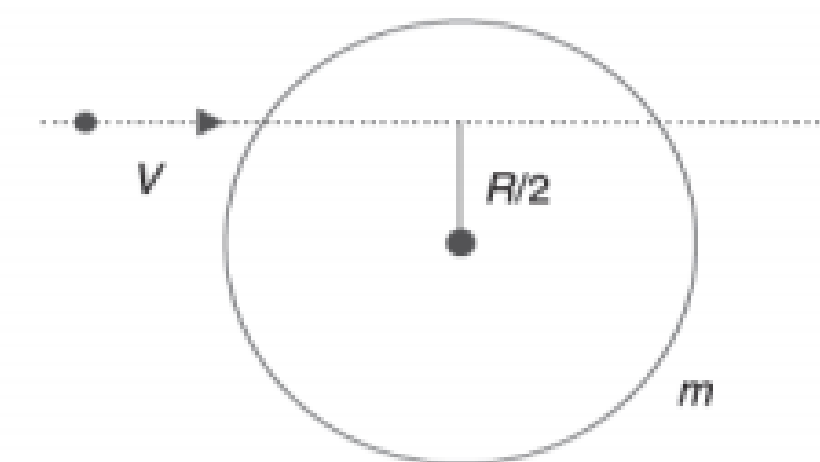
$$\frac{\omega l}{2} = \sqrt{5}V$$

A uniform rod of mass M is moving in a plane and has a kinetic energy of $\frac{4}{3}MV^2$ where V is speed of its centre of mass. Find the maximum and minimum possible speed of the end point of the rod.



$$e = \frac{(\omega R \cos 60^\circ + u_{cm})}{u} = \frac{\frac{m u}{2M} + \frac{m u}{M}}{\frac{3m u}{2M}} = e$$

A disc of mass m and radius R lies flat on a smooth horizontal table. A particle of mass m , moving horizontally along the table, strikes the disc with velocity V while moving along a line at a distance $\frac{R}{2}$ from the centre. Find the angular velocity acquired by the disc if the particle comes to rest after the impact.

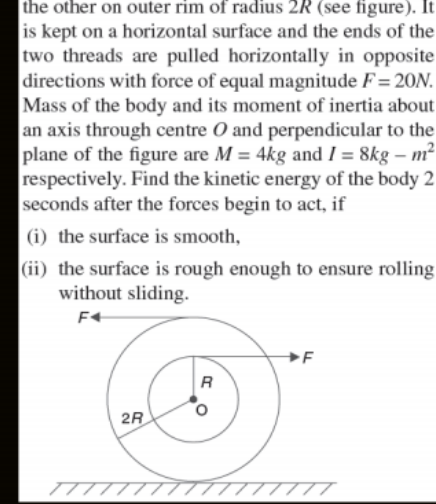


after collision find

① u_{cm} of disc

② ω of disc

and ③ e

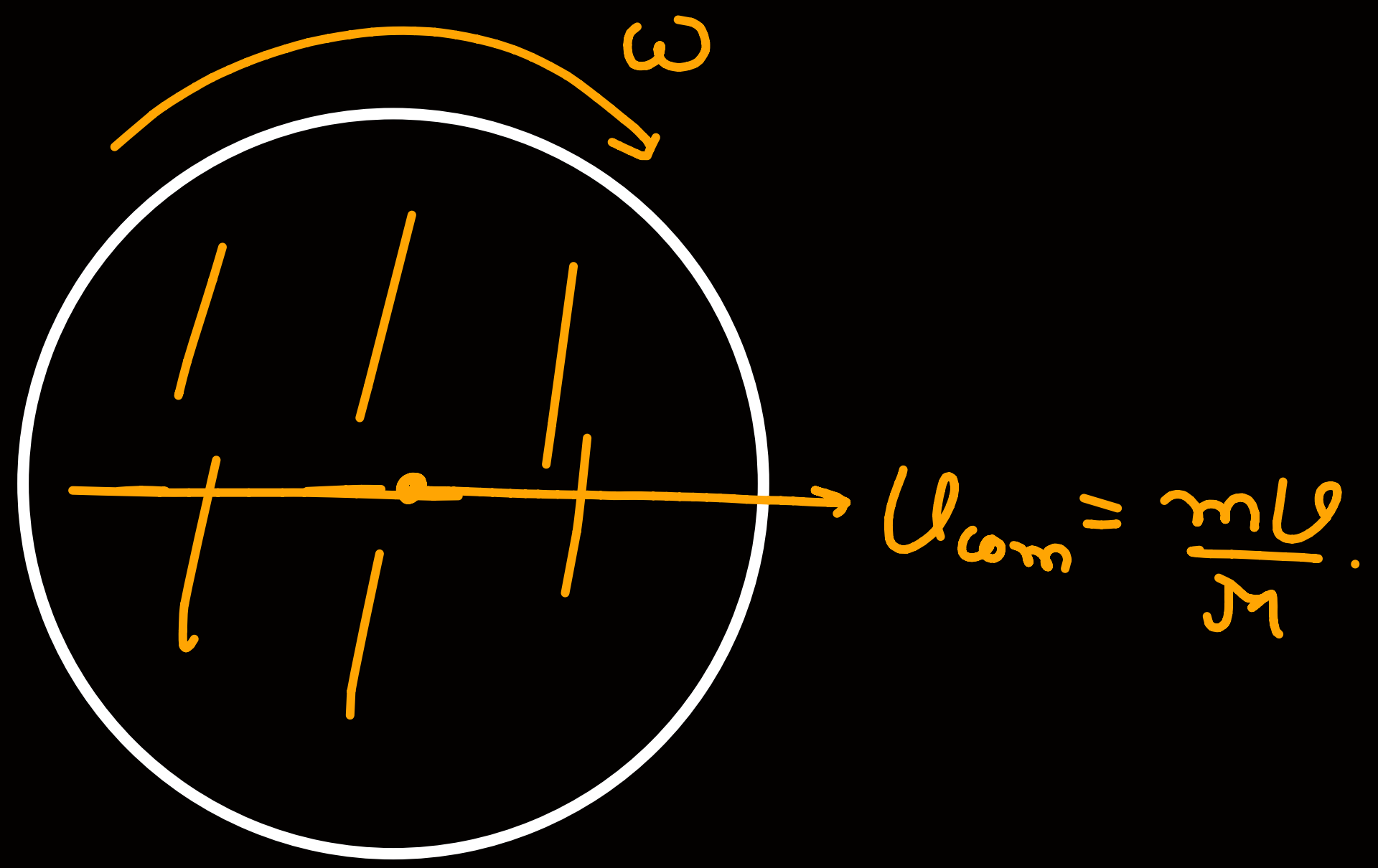


LMC

$$m v + 0 = 0 + M v_{com} \quad m \cdot$$

AMCT

$$m v \frac{R}{2} + 0 = 0 + 0 + \frac{M R^2}{2} \omega$$



$$v_{com} = \frac{m v}{M}$$

$$\omega = \frac{m v}{M R}$$

$$e = \underline{\hspace{2cm}}$$